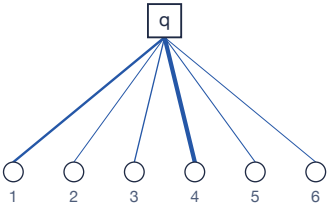
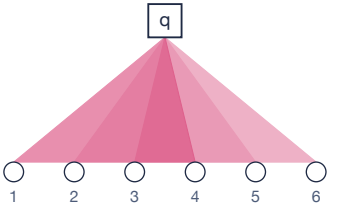
a Lookup: one score per token



score $Q_q \cdot K_j$ for each token j: n terms

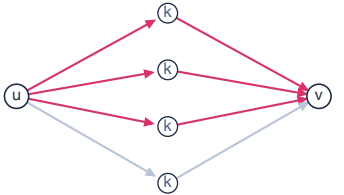
b Join: one score per pair



score $s_{qjk} = \sum_r Q_{qr} K_{jr} K'_{kr}$ for each pair:

n^2 terms (four of them shaded)

c Pair layer: a state per pair



state z_{uv} updated from (u, k) and (k, v)

for every middle k: two-step walks $u \to k \to v$